JAVA CHEAT SHEET — TESD 1400 CORE

BASIC STRUCTURE OF A JAVA PROGRAM

```
public class ClassName {
 public static void main(String[] args) {
 // your code here
 }
 }
```

Notes:

- File name must match class name (ClassName.java).
- main() is where the program starts running.

PRINTING OUTPUT

```
System.out.print("text"); // prints, no new line
 System.out.println("text"); // prints and moves to next line
 System.out.println(5 + 3); // you can print numbers and math
```

COMMENTS

Single-line comment:

```
 // this is a comment
```

Multi-line comment:

```
 /* this is
 a multi-line comment */
```

VARIABLES + DATA TYPES

```
int age = 19; // whole number
 double price = 3.99; // decimal number
 boolean done = true; // true or false
 char grade = 'A'; // single character
 String name = "Jeff"; // text (String is capitalized)
```

Final / constant:
final double TAX = 0.07; // cannot change later

BASIC MATH OPERATORS

- add
- subtract
- multiply
  / divide
  % remainder (modulo)

Examples:
int x = 10 % 3; // x = 1
double y = 7 / 2; // y = 3.0 if both are ints first (do 7.0/2 for 3.5)

INPUT FROM USER (SCANNER)

```
import java.util.Scanner;

Scanner input = new Scanner(System.in);

int n = input.nextInt(); // read int
 double d = input.nextDouble(); // read double
 String s = input.next(); // one word
 String line = input.nextLine(); // whole line of text
```

Example:

```
System.out.print("Enter age: ");
int age = input.nextInt();
```

STRING BASICS

```
String s = "hello";

s.length() // number of characters
 s.toUpperCase() // "HELLO"
 s.toLowerCase() // "hello"
 s.charAt(0) // 'h'
 s.equals("hello") // true/false, use this NOT ==
```

Don’t compare strings with ==
 Use .equals()

IF / ELSE (SELECTIONS)

```
if (condition) {
 // do this
 } else if (anotherCondition) {
 // do this
 } else {
 // fallback
 }
```

Relational operators:
 == equal
 != not equal

 greater than
 < less than
 = greater or equal
 <= less or equal

Logical operators:
 && and
 || or
 ! not

Example:

```
 if (score >= 90) {
 grade = 'A';
 } else if (score >= 80) {
 grade = 'B';
 } else {
 grade = 'C';
 }
```

SWITCH STATEMENT

```
switch (day) {
 case 1:
 System.out.println("Monday");
 break;
 case 2:
 System.out.println("Tuesday");
```

```
 break;
 default:
 System.out.println("Unknown");
 }
```

- break; is important, or it “falls through.”

WHILE LOOP

```
while (condition) {
 // repeat this while condition is true
 }
```

Example:

```
 int count = 0;
 while (count < 5) {
 System.out.println("Hi");
 count++;
 }
```

FOR LOOP

```
for (int i = 0; i < 5; i++) {
 System.out.println("Loop #" + i);
 }
```

Pattern:

```
 start ; condition ; update
```

DO-WHILE LOOP

```
do {
 // body runs at least once
 } while (condition);
```

Example:

```
 int n;
 do {
 System.out.print("Enter positive number: ");
 n = input.nextInt();
 } while (n <= 0);
```

RANDOM NUMBERS

```
import java.util.Random;

Random r = new Random();
 int x = r.nextInt(6); // 0 to 5
 int die = r.nextInt(6) + 1; // 1 to 6

double y = Math.random(); // 0.0 <= y < 1.0
 int pick = (int)(Math.random() * 10); // 0-9
```

METHODS (BASICS)

Define a method (function) outside main but inside the class:

```
public static int add(int a, int b) {
 return a + b;
 }
```

Call it:

```
 int sum = add(3, 4);
```

Void method (no return):

```
 public static void sayHi() {
 System.out.println("Hi");
 }
```

ARRAYS (1D)

```
int[] nums = new int[5]; // 5 spots, all 0
 nums[0] = 42; // set first
 int x = nums[0]; // read first

int[] grades = {90, 85, 70, 100}; // shortcut create
```

length:

```
 grades.length // number of elements
```

loop over array:

```
 for (int i = 0; i < grades.length; i++) {
 System.out.println(grades[i]);
 }
```

enhanced for loop (for-each):
```
for (int g : grades) {
System.out.println(g);
}
```

2D ARRAYS (BASICS)

```
int[][] grid = new int[3][3];
grid[0][1] = 7;
```

Loop through 2D:
```
for (int row = 0; row < grid.length; row++) {
for (int col = 0; col < grid[row].length; col++) {
System.out.print(grid[row][col] + " ");
}
System.out.println();
}
```

CASTING / TYPE CONVERSION

```
double d = 5; // int -> double is allowed
int n = (int)3.7; // force cut to 3

int n2 = Integer.parseInt("42"); // "42" → 42
double d2 = Double.parseDouble("2.5");// "2.5" → 2.5
```

MATH CLASS QUICK HITS

```
Math.pow(a, b) // a^b
Math.sqrt(x) // square root
Math.abs(x) // absolute value
Math.max(a, b) // bigger of two
Math.min(a, b) // smaller of two
Math.round(x) // rounds to nearest int
```

BOOLEAN LOGIC EXAMPLE

```
boolean ok = (age >= 18 && hasID);
if (ok) {
System.out.println("You may enter.");
} else {
```

```
System.out.println("Not allowed.");
}
```

COMMON ERROR REMINDERS

- Use .equals() to compare Strings, not ==
- Every { needs a matching }
- Arrays start at index 0, not 1
- Integer division drops decimals (7 / 2 == 3)
- Scanner nextInt() leaves the newline, so sometimes you need an extra nextLine() after it before reading text

MINI PATTERNS YOU REUSE A LOT

Print a menu and get choice:

```
System.out.println("1) Start\n2) Quit");
int choice = input.nextInt();
```

Sum loop:

```
int sum = 0;
for (int i = 0; i < nums.length; i++) {
sum += nums[i];
}
```

Validate input:

```
while (num < 0) {
System.out.println("Enter again:");
num = input.nextInt();
}
```

Method template:

```
public static returnType name(params) {
// work
return value;
}
```